
Knowledge is Overrated: A zero-knowledge machine learning and cryptographic hashing-based framework for verifiable, low latency inference at the LHC

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Abstract

1 Low latency event-selection (trigger) algorithms are essential components of Large
2 Hadron Collider (LHC) operation. Modern machine learning (ML) models have
3 shown great offline performance as classifiers and could improve trigger perfor-
4 mance, thereby improving downstream physics analyses. However, inference on
5 such large models does not satisfy the 40MHz online latency constraint at the LHC.
6 In this work, we propose a novel, theoretical framework built on cryptographic
7 techniques like hashing and zero-knowledge machine learning (zkML) to achieve
8 low latency inference, via a certifiable, early-exit mechanism from an arbitrarily
9 large baseline model. We lay the foundations for such a framework to achieve
10 nanosecond-order latency and discuss its inherent advantages, such as built-in
11 anomaly detection, within the scope of LHC triggers, as well as its potential to
12 enable a dynamic low-level trigger in the future.

13 1 Introduction

14 Accelerating ML inference to satisfy the LHC’s operational constraints is currently approached as an
15 ad-hoc optimization problem driven by an accuracy-acceleration tradeoff. Algorithms like AXOL1TL
16 and CICADA, recently tested [1] by the CMS experiment, combine this type of an approach with
17 hardware-specific optimization techniques, using frameworks like hls4ml [2]. These algorithms
18 successfully achieve $O(\text{ns})$ latencies for anomaly detection triggers [1]. However, we hypothesize
19 that, to achieve $O(\text{ns})$ latencies for a larger set of ML-based trigger decisions, we need a novel
20 approach to trigger design compatible with dynamic performance-improvement strategies.

21 We start with a short primer on polynomial fingerprinting, zkML, and early-exiting (EE) that form
22 the building blocks of our framework.

23 1.1 Polynomial fingerprinting

24 **Hash function:** Any function that maps a mathematical object, like a vector of arbitrary size, to a
25 fixed-size value [3, 4]. We restrict the definition to non-cryptographic hash functions [5] since we do
26 not require security-related properties in the hashing algorithm for our framework. Say we have a
27 family of hash functions $\mathcal{H}\{h : \mathcal{M} \rightarrow \mathcal{Y}\}$. We can then define a probabilistic hashing algorithm as a
28 scheme, that for a given arbitrary-size object $M_i \in \mathcal{M}$, chooses a random hash function $h_i \in \mathcal{H}$ to
29 evaluate the hash.

30 For a probabilistic hash to serve as a near-unique fingerprint for all objects in $\mathcal{M}$ we require that the
31 collision probability [4] over a random h_i , for any 2 arbitrary values $x, y \in \mathcal{M}$, be bounded by a
32 small, well-defined value ϵ , written as,

$$\Pr_{h_i \in \mathcal{H}} [h_i(x) = h_i(y)] \leq \epsilon \quad (1)$$

Rabin fingerprinting [6]: A hash defined over a finite, Galois field, $\text{GF}(2)$, that maps an n -bit vector represented as a polynomial (using techniques like polynomial interpolation [7]), $P(x)$ of degree $(n - 1)$, to the remainder, $r(x)$, obtained on dividing $P(x)$ by $p(x)$; where $p(x)$ is an irreducible polynomial of degree k over $\text{GF}(2)$ [6]. Thereby the Rabin Fingerprinting algorithm reduces a large n -bit vector to a unique k -bit fingerprint.

For our case, let's define the Rabin fingerprint over a large finite field of prime order, $\mathbb{F}_p$, as,

$$h = P(x) \pmod{p} \quad (2)$$

The collision probability is then described by the Schwartz-Zippel Lemma [8]. The lemma states that for any two distinct polynomials $P_1(x), P_2(x)$, each of degree at most d , then the collision probability for a randomly chosen point $q \in \mathbb{F}_p$ is given as,

$$\Pr_{q \in \mathbb{F}_p} [P_1(q) = P_2(q)] \leq \frac{d}{|\mathbb{F}_p|} \quad (3)$$

Thereby, choosing a finite field ordered by a large enough prime, and constraining the highest polynomial degree to be low, leads to statistically negligible collision probabilities.

1.2 Zero-knowledge machine learning

Zero-Knowledge Proof: A ZKP is a cryptographic protocol that allows one party, the prover, to convince another party, the verifier, that a given statement is true, without revealing any information beyond the fact of the statement's truth [9]. A powerful application of modern ZKP systems lies in their ability to guarantee computational integrity, making them tools for fast verifiability. The core properties of ZKPs are completeness, soundness and privacy-preservation.

zkML is an application of ZKPs for generating a succinct proof, π , of the correct execution of an ML model's inference computation, $C(x, w) = y$, where $x \in L$ is a public input in a corpus L (e.g., training dataset), w is the private witness (e.g., model parameters, intermediate activations) and y is the public model output. For our usage, privacy is not a major factor and we focus on verifiability, which for a given prover $\mathcal{P}$ and verifier $\mathcal{V}$ is defined, as in [10], by the completeness,

$$\Pr[\langle \mathcal{P}(x, w) \leftrightarrow \mathcal{V}(x) \rangle = \text{accept}] = 1 \quad (4)$$

and the soundness, for $x' \notin L$ and a random prover $\mathcal{P}^*$,

$$\Pr[\langle \mathcal{P}^*(x', w) \leftrightarrow \mathcal{V}(x') \rangle = \text{accept}] \leq \epsilon \quad (5)$$

where $\langle \rangle$ represents prover-verifier interactions.

The foundational step of most zkML systems is arithmetization; the process of transforming computation C into a family of algebraic equations. These equations serve as polynomial constraints that are tractable when defined over a finite field $\mathbb{F}_p$, and the validity of the computation is reduced to proving that these constraints hold. The complexity of a zkML system is largely concentrated on the arithmetization stage. The subsequent steps involve using cryptographic primitives such as hashing and Polynomial Commitment Schemes (PCS) [9] to compress the constraints down to unique, tractable commitments (e.g., fingerprints), used for faster verification. The Schwartz-Zippel lemma (Eq.3) again serves as a bound for the integrity of the proof.

We focus on a specific type of zkML system based on zk-STARK (Zero-Knowledge Scalable Transparent ARguments of Knowledge) [11]. zk-STARKs are typically constructed under minimal cryptographic assumptions which gives rise to scalability and transparency. The complexity, for a given computation of size $T(n)$, scales quasi-linearly, $O(T(n)\text{poly log } T(n))$ for proof generation and poly-logarithmically, $O(\text{poly log } T(n))$ for proof verification [11]. The transparency makes the system auditable and reproducible.

71 1.3 Early-Exit

72 EE [12] strategies are a popular approach for significantly reducing inference time and computation of
 73 AI algorithms running on low power edge-devices. A typical implementation of this method involves
 74 multiple exit-branches at different stages of the forward pass of a large ML model, where each branch
 75 is composed of a smaller classifier [13], with preset exit-logic. While this approach significantly
 76 reduces computation for some online samples, multiple exit-branches and complex exit-logic likely
 77 do not satisfy LHC trigger latency constraints [13].

78 2 PHAZE: Probabilistic Hashing And zkML-based Early-exit

79 We present PHAZE, a novel EE framework designed for a new generation of low latency ML-based
 80 triggers, built with the cryptographic primitives described in Sec.1. For the LHC, feasibility studies
 81 for such a framework should broadly cover: physics performance in terms of trigger efficiency related
 82 parameters; verifiability for data quality monitoring and reproducing trigger decisions for downstream
 83 analyses; operational constraints laid by hardware resources; computational constraints defined by
 84 the accelerator and detectors.

85 **Scope and assumptions:** In this work we restrict ourselves to accelerating the inference process of
 86 a performant ML classifier that acts as the main triggering algorithm with the following assumptions:

- 87 1. Baseline model: We assume its availability and design choices for this large, high-accuracy
 88 model, $\mathbb{M}_{\text{full}}$, like a single, large, general purpose physics model versus multiple small
 89 specialized models; the model architecture etc., are left out of our scope.
- 90 2. Representative dataset: We assume that a large dataset corresponding to the input-space $\mathcal{I}$,
 91 is available to train on and serves as an acceptable representation of multiple classes of real
 92 events (signals, backgrounds and noise) commonly encountered at the LHC.
- 93 3. Early predictive sufficiency: We assume that early-layer activations, $\mathcal{A} \in \mathbb{R}^k$ of $\mathbb{M}_{\text{early}} \subset$
 94 $\mathbb{M}_{\text{full}}$, act as EE signals and are a strong proxy to the model decisions $\mathcal{D} = \{0, 1\}^t$ (for
 95 t different trigger classes), for a large enough subset $I \subseteq \mathcal{I}$. Here $\mathbb{R}^k$ is a k -dimensional
 96 real, latent activation vector space such that $\mathbb{M}_{\text{early}} : I \rightarrow \mathbb{R}^k$. Design choices to fulfill this
 97 requirement may involve knowledge distillation (KD), fine-tuned adapter layers on top of a
 98 pre-trained $\mathbb{M}_{\text{full}}$ or other such strategies and this choice is out of scope for our work.
- 99 4. Injectivity of polynomial interpolation: We assume that a polynomial interpolation scheme
 100 that maps a vector $A^* \in \mathbb{R}^k$ to a polynomial $P_{A^*}(x)$ is available and injective. Here, $P_{A^*}(x)$
 101 belongs to the space $\mathcal{P}_d(\mathbb{F}_p^m)$ of all m -variate polynomials of degree at most d .

102 2.1 Method

103 PHAZE is designed to be a two-stage algorithm: a computation-intensive offline phase that is not
 104 constrained by latency requirements or hardware choices and a low latency online phase that is
 105 expected to be a part of the online trigger algorithm.

106 **Offline phase:** We start by training the large baseline model to obtain the mapping $\mathbb{M}_{\text{full}} : \mathcal{I} \rightarrow \mathcal{D}$,
 107 to satisfactory classification performance. Based on the assumption above, this trained model can be
 108 approximated by the map EE: $\mathcal{A} \rightarrow \mathcal{D}$, defined as the activation extraction step. While this EE can
 109 be statistically equivalent to KD, PHAZE does not smear over unseen events, and that is essential for
 110 discoveries. Each activation $A_j \in \mathcal{A}$ corresponding to $I_j \in \mathcal{I}$ is quantized to $A_j^* \in \mathbb{F}_p^k$. The choice
 111 between quantization during versus after training is left to experimental evaluation. We then generate
 112 an injective map for each A_j^* to polynomial $P_{A_j^*}$ that passes through a predefined set of k evaluation
 113 nodes, $\{x_1, \dots, x_k\} \subset \mathbb{F}_p^m$ such that $P_{A_j^*}(x_i) = A_{j,i}^*$ for all $i \in \{1, \dots, k\}$. A rabin fingerprint is then
 114 generated as a hash for each such $P_{A_j^*}$ as described in Eq.2 with each polynomial evaluated at a
 115 constant, random challenge point $r = (r_1, \dots, r_m)$ sampled from $\mathbb{F}_p^m$. This fingerprint h_j is then used
 116 as a key to populate a decision map, the Verifiable Decision Map (VDM), whose value is $D_j \in \mathcal{D}$.
 117 Next, we start the zk-STARK based proof generation, by reusing the polynomial interpolation and
 118 hashing steps. The computational integrity (CI) statement to be proven is formulated as a compound
 119 assertion S_j for each I_j given by:

120 *"The inference of $\mathbb{M}_{\text{full}}$ on I_j yields D_j AND inference of $\mathbb{M}_{\text{full}}$ on I_j yields a*
 121 *quantized A_j^* , whose canonical polynomial $P_{A_j^*}$ evaluates to the hash value h_j at*
 122 *the public challenge point r ."*

123 This statement is converted into an Algebraic Intermediate Representation (AIR), which consists of
 124 an execution trace of the whole system with corresponding polynomial constraints that hold if and
 125 only if the computations were executed correctly as explained in Sec.1.2. The zk-STARK prover
 126 $\mathcal{P}$ then uses the AIR to generate a succinct proof π_j to certify S_j . As discussed above, this proof
 127 generation, though computation-intensive, scales to reasonably large models, with the proofs being
 128 transparent for ad-hoc downstream auditing and verification. π_j now certifies the validity of the
 129 decision map $\text{VDM}: h_j \rightarrow D_j$. The proofs π_j make the decision map resistant to corruptions such
 130 as accidental bit-flips and can then be archived.

131 **Online phase:** This phase is deployed at the trigger level and it starts with the same activation
 132 extraction step as in the offline phase to produce A_{new}^* for each new event I_{new} , ideally optimized
 133 for specialized hardware like FPGAs. Dedicated hardware cores then perform an on-the-fly (OTF)
 134 hashing step to directly evaluate $h_{\text{new}} = P_{A_{\text{new}}^*}(r)$ using the numerically stable Barycentric Lagrange
 135 interpolation and evaluation method [14] which skips a major portion of the polynomial definition in
 136 the offline phase since we do not need to generate a new proof online. Then a fast VDM lookup is
 137 performed using h_{new} as the key to retrieve the trigger decision, $D' = \text{VDM}[h_{\text{new}}]$.

138 A map-miss occurs when no VDM key matches h_{new} . This could indicate either a physics anomaly
 139 or an unknown detector anomaly. In either case, PHAZE inherently serves as an anomaly detector,
 140 with potential for incorporating specialized, downstream anomaly handling algorithms by buffering
 141 such events or by setting a default action (e.g., discard the event).

Table 1: Breakdown of computational tasks in the PHAZE framework.

Phase	Task	Latency per event	Computational complexity
Offline	Full Model Inference	$O(\text{ms} - \text{s})$	$O(\mathbb{M}_{\text{full}})$
	EE Activation Extraction	$O(\mu\text{s})$	$O(\mathbb{M}_{\text{early}})$
	Polynomial Interpolation	$O(\text{ms} - \text{s})$	$O(\text{poly}(k \log k))$
	Probabilistic Hashing	$O(\mu\text{s})$	$O(d)$
	zk-STARK Proof Generation	$O(\text{min})$	$O(\mathbb{M}_{\text{full}} \text{poly-log}(\mathbb{M}_{\text{full}}))$
	ad-hoc Proof Verification	$O(\mu\text{s} - \text{ms})$	$O(\text{poly-log}(\mathbb{M}_{\text{full}}))$
	VDM Population	$O(\text{ns})$	$O(1)$ per event
Online	EE Activation Extraction	$O(\text{ns})$ (FPGA)	$O(\mathbb{M}_{\text{early}})$
	On-the-Fly Hashing	$O(\text{ns})$ (FPGA)	$O(\sqrt{d})$ Erstin's method
	VDM Lookup	$O(\text{ns})$ (FPGA)	$O(1)$ from BRAM

142 **Latency estimates per event:** The net theoretical latency for the online trigger is thereby given
 143 by $T_{\text{online}} = T_{\text{EE}} + T_{\text{OTF-hashing}} + T_{\text{VDM-lookup}}$. A conservative estimate for T_{EE} , for an optimized
 144 $\mathbb{M}_{\text{early}}$ on an FPGA would be 100-200ns [15, 16]. OTF-hashing using fast polynomial evaluation (e.g.
 145 Estrin's method [17]) takes as low as ~ 20 clock cycles on modern FPGAs, giving $T_{\text{OTF-hashing}} \approx 50\text{ns}$,
 146 while a VDM lookup can be performed in a single clock cycle, giving $T_{\text{VDM-lookup}} \approx 2.5\text{ns}$. Thereby
 147 a conservative estimate of T_{online} for a single event across a reasonable number of trigger classes
 148 would be in the range of 152.5-252.5ns on modern FPGAs.

149 3 Conclusions

150 We present the PHAZE theoretical framework to potentially achieve nanosecond-order trigger latencies
 151 at the LHC using foundation model-scale ML algorithms, under a set of peripheral assumptions. To
 152 the best of our knowledge, this is the first physics framework to utilize cryptographic primitives to
 153 enable fast inference and ensure verifiability of trigger algorithms, which is a key step in the direction
 154 of a novel, dynamically adaptive event selection paradigm at the LHC. We include further comments
 155 on feasibility in the appendices and the current assumptions are part of our future work in realizing
 156 PHAZE as an experimental proof-of-concept system. PHAZE is also compatible as a building block for
 157 a larger distributed algorithm that is dynamically updated, while remaining verifiable, for flexible low
 158 level selections as a new paradigm for trigger design (see appendix C).

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A Probabilistic guarantees and error components

The model accuracy $\varepsilon(\mathbb{M}_{\text{full}})$ is a major component of the net achievable trigger efficiency for a framework like PHAZE, and is framework-independent by assumption. The key component leading to a loss in $\varepsilon(\mathbb{M}_{\text{full}})$ is the probability of predictive insufficiency for the map EE: $a_j \in \mathcal{A} \rightarrow d' \in \mathcal{D}$, given by $\Pr[d' \neq d_j]$, for $d_j = \mathbb{M}_{\text{full}}[I_j]$. The operational bounds for this term depend on the EE strategy chosen and can be statistically estimated from experimental validation. The quantization of the latent activation vector can introduce prediction errors as well. However, this can be mitigated to an extent using representation learning techniques, like contrastive loss terms, that force the early layers to place different classes in distinct regions of the latent activation space. Events on the boundaries of these regions would have a higher chance of misclassification and thereby their

contributions are encapsulated in the accuracy losses above. The hashing itself introduces an error when a collision occurs. However, for a 64-bit hash and an expressive polynomial degree choice of $d = 100$, the upper bound given by Eq.3 is $100/2^{64}$, which is computationally insignificant.

B Hardware resource feasibility

The main accelerating factor that PHAZE takes advantage of is moving most of the heavy computation to the offline phase where we are not compute/latency constrained. For the online phase, evaluating $EE: \mathcal{A} \rightarrow \mathcal{D}$ has shown to be feasible for reasonably large models using `hls4ml`. OTF-hashing using optimizations on Horner’s rule or Estrin’s method are viable on modern FPGAs (e.g., Xilinx Virtex 6) [18]. The VDM itself is a limiting component since its most efficient use requires storage in FPGA memory. For e.g., AMD UltraScale+ FPGA series provides 455Mb memory. For a 64-bit hash, each VDM entry would require say 72 bits, for the hash and the decision across say 8 trigger classes. This means a single FPGA can hold evaluations for $\frac{455 \times 10^6}{72} \approx 6.3 \times 10^6$ events. This size is not sufficient for a representative dataset and thereby strategies to enable fast distributed look-ups across multiple VDMs would be preferable. This is currently an open research question we look to answer as part of our future work.

C Inherent capabilities and future directions

C.1 Map-miss anomaly detection

Map-misses are an intrinsic tool that can provide "computationally free", low-level anomaly detection. Map-miss rates are strongly tied to representative dataset description, EE prediction sufficiency and VDM size and can be counter-productive if the above assumptions are not met. However, map-misses can strongly point to anomalous physics events or novel, unknown detector effects, that would otherwise be missed by current trigger algorithms. Such events would generally be discarded, or would require dedicated anomaly detection algorithms at the low-level trigger, which increases the computational overhead. Both these types of anomalies are essential to analyze for better long-term physics, as well as detector, performance. Leveraging map-misses in PHAZE-like trigger paradigms and combining it with powerful downstream trigger analyses or a dynamic VDM setup could potentially improve chances for discovery.

C.2 Dynamic VDM

The initial proposal envisions a static VDM, compiled once from a fixed dataset. However, the architecture readily supports dynamic updates, allowing the trigger system to adapt over time without requiring a full hardware redesign or resynthesis. The zkML prover-verifier setup is crucial to maintain the integrity of such a system.

Scheduled updates: The most direct approach involves periodically updating the VDM during planned LHC operational pauses, such as technical stops or shutdowns. During these periods, a new, more comprehensive VDM can be compiled offline. This updated map could incorporate knowledge gained from previous runs, adapt to changes in detector calibration and conditions, or, most significantly, include newly understood event topologies derived from classes of events with high map-miss rates.

Autonomous updates: A more ambitious form of a dynamic VDM would involve using knowledge accumulated online by downstream high-level triggers (HLT), and intermittently use beam downtime to update the VDM. This would require smaller baseline models that can simulate artificial maps between the event hashes and the HLT trigger decisions for event topologies with high map-miss rates. However, this would also require very slow online proof generation before the VDM can be altered. The current technology gap to realize a system like this is high, however, an autonomous trigger system can greatly improve physics analyses, specially in the direction of new physics searches.